



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester – Fall 2021

Paper: Classical Mechanics

Course Code: PHY-301

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Roll No.
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Time: 3 Hrs. Marks: 6

Q.1 Solve the following:

(6x5=30)

1. Show that the total angular momentum of a system of particles is equal to the sum of angular momentum of center of mass and the angular momentum of the particles about the center of mass.
2. State D'Alembert's Principle and use it to derive the Lagrange's equation of motion

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} \right) - \frac{\partial T}{\partial q_i} = Q_i.$$

3. The spherical pendulum has Lagrangian

$$L = \frac{1}{2} m l^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) - m g l \cos \theta$$

Write down the Euler-Lagrange equations of motion.

4. Show that the Poisson bracket of two integrals of motion is again an integral of motion.
5. A sphere of radius ρ is constrained to roll without slipping on the lower half of the inner surface of a hollow cylinder of inside radius R . Determine the Lagrange function, the equation of constraint and Lagrange's equation of motion. Find the frequency of small oscillation.
6. If L is a Lagrangian for a system of n degrees of freedom satisfying Lagrange equation of motion, show by direct substitution that

$$L' = L + \frac{d}{dt} F(q_1, \dots, q_n; t),$$

also satisfies the Lagrange's equation of motion where F is an arbitrary differentiable function of its argument.

Q.2. Solve the following.

(3x10=30)

1. Show that the Poisson bracket satisfies the Jacobi identity $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$. (10)
2. Consider the motion of a particle in a central force field

$$V(r) = -\frac{k}{r}.$$

Write down the Lagrangian in polar coordinates and integrate the equation of motion to derive

$$\theta(r) = \int \frac{l \, dr}{r^2 \sqrt{2\mu \left(E + \frac{k}{r} - \frac{l^2}{2\mu r^2} \right)}} + \text{constant},$$

where E is the total energy and l is the angular momentum. Now change variables as $u = \frac{l}{r}$ to derive the equation of a conic section

$$\frac{\alpha}{r} = 1 + \epsilon \cos \theta.$$

3. Show that the transformation

$$P = q \cot p,$$

$$Q = \ln \left(\frac{\sin p}{q} \right),$$

is canonical and also show that the corresponding generating function is

$$F = e^{-Q} (1 - q^2 e^{2Q})^{\frac{1}{2}} + q \sin^{-1} (q e^Q)$$

This law holds in the absence of external torque and validity of action & reaction law.

Consider the angular momentum of a system with respect to an origin (O) (is reference point). When the system is at rest the "angular momentum is described as cross-product of $\vec{r}_i \times p_i$ and summing it all value of 'i'.

$$\vec{P}_L = \vec{F}_i(e) + \sum_i \vec{F}_{ji}$$

$$\text{or } \frac{d}{dt} P_L = "$$

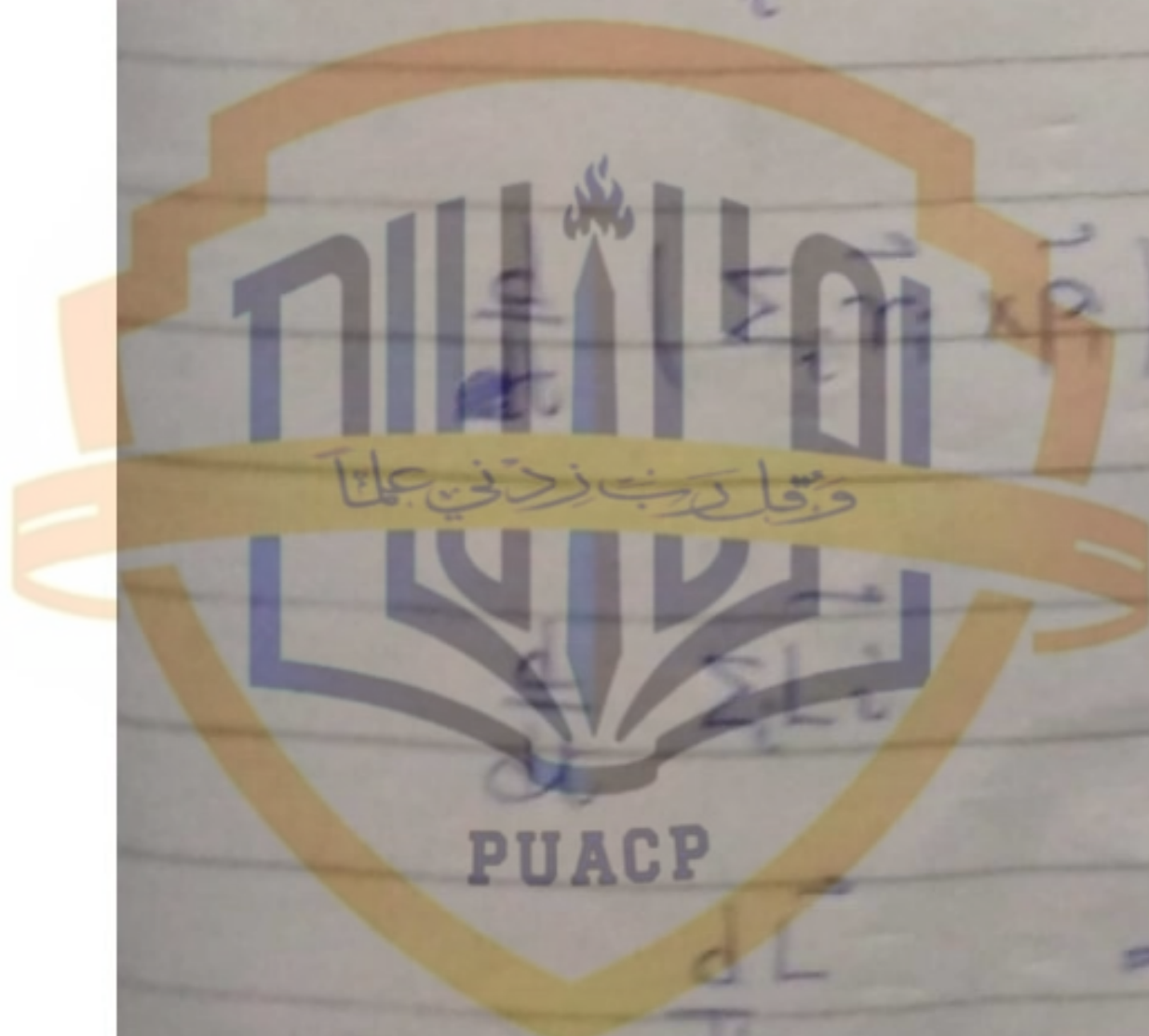
Taking cross-product with $\vec{r}_i$ and summing over all 'i' gives:

$$\sum_i \vec{r}_i \times \frac{d}{dt} \vec{P}_L = \sum_i \vec{r}_i \times \vec{F}_i(e) + \sum_{\substack{i,j \\ j \neq i}} \vec{r}_i \times \vec{F}_{ji}$$

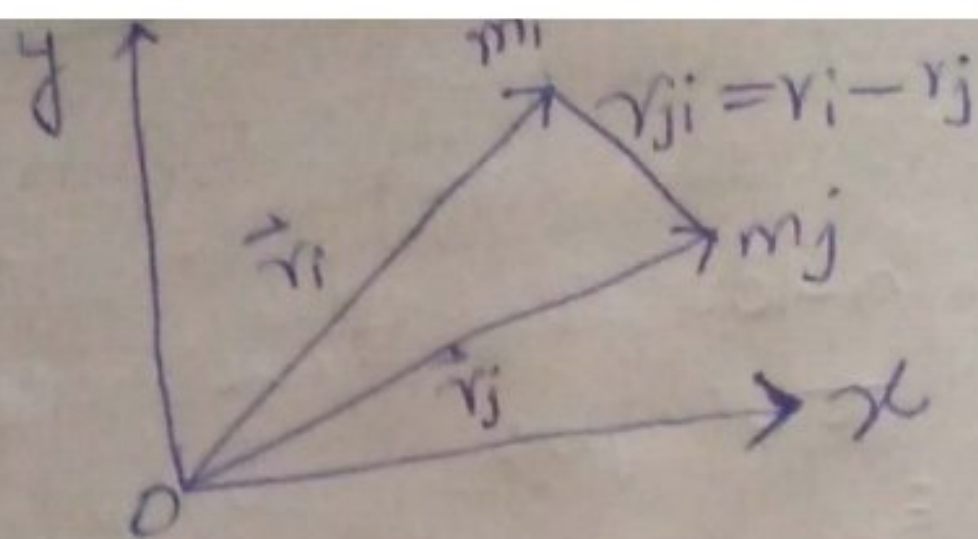
$$\frac{d}{dt} \left(\sum_i \vec{r}_i \times \vec{P}_L \right) = "$$

$$= "$$

$$\frac{dL}{dt} = \sum_i \vec{r}_i \times \vec{F}_i(e) + \sum_{\substack{i,j \\ j \neq i}} \vec{r}_i \times \vec{F}_{ji} \rightarrow (i)$$



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Taking 'term' $\sum_{i,j} \vec{r}_i \times \vec{F}_{ji}$

$$\sum_{i,j} \vec{r}_i \times \vec{F}_{ji} = \vec{r}_i \times \vec{F}_{ji} + \vec{r}_j \times \vec{F}_{ij} \rightarrow (ii)$$

By Newton's law of action & reaction

i.e. $\boxed{\vec{F}_{ij} = -\vec{F}_{ji}}$

$$\text{So } \sum_{i,j} \vec{r}_i \times \vec{F}_{ji} = \vec{r}_i \times \vec{F}_{ji} - \vec{r}_j \times \vec{F}_{ji}$$

$$= (\vec{r}_i - \vec{r}_j) \times \vec{F}_{ji} \rightarrow (ii)$$

$$(\vec{r}_i - \vec{r}_j) \times \vec{F}_{ji} \rightarrow (iii)$$

From fig 3

$$\vec{r}_{ji} = \vec{r}_j + \vec{r}_{ji}$$

$$\text{or } \vec{r}_i - \vec{r}_j = \vec{r}_{ji}$$

Where $\vec{r}_{ji}$ is
from i to
and $\vec{F}_{ji}$ is
acting along
these

a vector
is particle
a force
the line
Particle

here

$$\vec{r}_{ji} \times \vec{F}_{ji} =$$

and $\sum_{i,j} \vec{r}_i \times \vec{F}_{ji}$

Now eq

$$\frac{d\vec{L}}{dt} =$$

$$\frac{d\vec{L}}{dt} =$$

$$\text{If } \vec{r} =$$

$$\Rightarrow \vec{L}$$

If no

On the
momentum
remain con
law of

here

$$\vec{r}_{ji} \times \vec{F}_{ji} = r_{ji} F_{ji} \sin \theta$$

$$= r_{ji} F_{ji} \sin 0^\circ$$

$$\because \sin 0^\circ = 0$$

$$= 0$$

$$\text{and } \sum_{\substack{ij \\ i \neq j}} \vec{r}_i \times \vec{F}_{ji} = 0$$

Now eq (i) becomes

$$\frac{d\vec{L}}{dt} = \sum_i \vec{r}_i \times \vec{F}_i^{(e)} + 0$$

$$= \sum_i \vec{r}_i \times \vec{F}_i^{(e)}$$

$$= \sum_i \vec{\tau}_i^{(e)}$$

$$\frac{d\vec{L}}{dt} = \vec{\tau}^{(e)}$$

$$\text{If } \vec{\tau}^{(e)} = 0, \text{ then } \frac{d\vec{L}}{dt} = 0$$

$$\Rightarrow \vec{L} = \text{Constant}$$

If no external torque acting on the system, the angular momentum of the system of particles remain constant, thus it gives conservation of momentum.

Langrangians equations-

The langrange equation is the mathematical eq. that describes the motion of system of particles. It is named after the mathematician J.N Langrange, who introduced it in 1788.

It is a set of 2nd order differential equation for a system of particles which relates the k.E (T) of the system to the generalize co-ordinates (q_i), the generalize force (Q_i) and time (t).

The langrange eq. provide an easy way to solve complex mechanics which are difficult to solve.

i) Lagrange's eq for the dynamic system.

From De Alembert's principle,

$$\sum_i (\vec{F}_i^{(c)} - \vec{p}_i) \cdot \delta \vec{r}_i = 0$$

$$\Rightarrow \sum_i \vec{F}_i^{(c)} \cdot \delta \vec{r}_i - \sum_i \vec{p}_i \cdot \delta \vec{r}_i = 0$$

$$\Rightarrow \sum_i \vec{F}_i^{(c)} \cdot \delta \vec{r}_i = \sum_i \vec{p}_i \cdot \delta \vec{r}_i \rightarrow (i)$$

Now we take;

$$\vec{p}_i = \vec{F}_i = M_i \vec{a}_i = M_i \frac{d^2 \vec{r}_i}{dt^2}$$

Where M_i is mass of the i th particle, $\vec{r}_i$ is its position vector

By def. of virtual displacement (in terms of generalized co-ordinates)

$$\delta \vec{r}_i = \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \cdot \delta q_j$$

Now right hand of eq (i) becomes

$$\sum_i \vec{p}_i \cdot \delta \vec{r}_i = \sum_i M_i \vec{r}_i \cdot \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \cdot \delta q_j$$

$$\parallel = \sum_j \left(\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \right) \delta q_j$$

(1)

Langrangian equation: 31/01/2024

$$\sum_i \vec{F}_i^{(c)} \cdot \delta \vec{r}_i = \sum_i \vec{p}_i \cdot \delta \vec{r}_i \rightarrow (i)$$

$$\sum_i \vec{p}_i \cdot \delta \vec{r}_i = \sum_j \left(\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \right) \delta q_j \rightarrow (ii)$$

Consider relation

$$\frac{d}{dt} \left(M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \right) = \sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} +$$

$$\sum_i M_i \vec{r}_i \cdot \frac{d}{dt} \left(\frac{\partial \vec{r}_i}{\partial q_j} \right)$$

$$\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} = \sum_i \frac{d}{dt} \left(M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \right) -$$

$$\sum_i M_i \vec{r}_i \cdot \frac{d}{dt} \left(\frac{\partial \vec{r}_i}{\partial q_j} \right)$$

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Exp. of velocity in terms of generalized velocity.

$$\vec{r}_i = \sum_j \frac{\partial \vec{r}_i}{\partial \dot{q}_j} \dot{q}_j + \frac{\partial \vec{r}_i}{\partial t} = \vec{v}_i$$

Differentiating w.r.t. $\dot{q}_j$, we get

$$\frac{\partial \vec{r}_i}{\partial \dot{q}_j} = \frac{\partial}{\partial \dot{q}_j} \left(\sum_j \frac{\partial \vec{r}_i}{\partial \dot{q}_j} \dot{q}_j + \frac{\partial \vec{r}_i}{\partial t} \right)$$

$$\text{or } \frac{\partial \vec{v}_i}{\partial \dot{q}_j} = \sum_j \frac{\partial \vec{r}_i}{\partial \dot{q}_j} \cdot \frac{\partial \dot{q}_j}{\partial \dot{q}_j} + \frac{\partial}{\partial \dot{q}_j} \left(\frac{\partial \vec{r}_i}{\partial t} \right)$$

$$\frac{\partial \vec{v}_i}{\partial \dot{q}_j} = \sum_j \frac{\partial \vec{r}_i}{\partial \dot{q}_j} \cdot (1) + 0 = \sum_j \frac{\partial \vec{r}_i}{\partial \dot{q}_j}$$

From Calculus

$$\frac{\partial \vec{v}_i}{\partial \dot{q}_j} = \frac{\partial}{\partial \dot{q}_j} \left(\frac{d\vec{r}_i}{dt} \right) = \frac{d}{dt} \left(\frac{\partial \vec{r}_i}{\partial \dot{q}_j} \right)$$

(4)

Putting eq (3) & (4) in (2)

$$\sum_i M_i \vec{r}_i \frac{\partial \vec{r}_i}{\partial \dot{q}_j} = \sum_i \left[\frac{d}{dt} (M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_j}) - \right.$$

$$\left. M_i \vec{r}_i \cdot \frac{d}{dt} \left(\frac{\partial \vec{r}_i}{\partial \dot{q}_j} \right) \right]$$

$$\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_j} = \sum_i \left[\frac{d}{dt} (M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_j}) - M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_j} \right] \quad (5)$$

Now,

$$\frac{\partial}{\partial \dot{q}_j} (M_i V_i^2) = \frac{\partial}{\partial \dot{q}_j} (M_i \cdot \vec{V}_i \cdot \vec{V}_i) = M_i \left(\vec{V}_i \cdot \frac{\partial \vec{V}_i}{\partial \dot{q}_j} + \frac{\partial \vec{V}_i}{\partial \dot{q}_j} \cdot \vec{V}_i \right)$$

Taking Partial derivative

$$\text{or } \frac{\partial}{\partial \dot{q}_j} (M_i V_i^2) = 2 M_i \vec{V}_i \cdot \frac{\partial \vec{V}_i}{\partial \dot{q}_j}$$

$$M_i \vec{V}_i \cdot \frac{\partial \vec{V}_i}{\partial \dot{q}_j} = \frac{1}{2} \frac{\partial}{\partial \dot{q}_j} (M_i V_i^2) \quad (6)$$

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Simlarily,

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$$\frac{\partial}{\partial \dot{q}_j} (M_i V_i^2) = 2 M_i \vec{V}_i \cdot \frac{\partial \vec{V}_i}{\partial \dot{q}_j}$$

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$$\text{Or } M_i \vec{v}_i \cdot \frac{\partial \vec{v}_i}{\partial \dot{v}_j} = \frac{1}{2} \frac{\partial}{\partial \dot{v}_j} (M_i v_i^2)$$

(7)

Using eq (6) & (7) in (5)

$$\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{v}_j} = \sum_i \left[\frac{d}{dt} \left(\frac{1}{2} \frac{\partial}{\partial \dot{v}_j} (M_i v_i^2) \right) - \frac{1}{2} \frac{\partial}{\partial \dot{v}_j} (M_i v_i^2) \right]$$

$$\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{v}_j} = \sum_i \left[\frac{d}{dt} \left(\frac{\partial}{\partial \dot{v}_j} \left(\frac{1}{2} M_i v_i^2 \right) - \frac{\partial}{\partial \dot{v}_j} \left(\frac{1}{2} M_i v_i^2 \right) \right) \right]$$

$$\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{v}_j} = \left[\frac{d}{dt} \left(\frac{\partial}{\partial \dot{v}_j} \left(\sum_i \frac{1}{2} M_i v_i^2 \right) - \frac{\partial}{\partial \dot{v}_j} \left(\sum_i \frac{1}{2} M_i v_i^2 \right) \right) \right]$$

$$\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_j} = \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \right]$$

$$T = \sum_i \frac{1}{2} M_i \dot{r}_i^2$$

Multiply this eq both sides by δq_j and then making summation

$$\sum_j \left(\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_j} \right) \delta q_j = \sum_j \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \right] \delta q_j$$

$$\text{or } \sum_j \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \right] \delta q_j = \sum_j \left[\sum_i M_i \vec{r}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_j} \right] \delta q_j$$

Comparing (1) & (2)

(8)

$$\sum_j \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \right] \delta q_j = \sum_i \vec{p}_i \cdot \delta \vec{r}_i$$

Also definition $\delta W = \sum_j Q_j \delta q_j$

Where Q_j is the generalized force

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These are equations of motion for a spring pendulum.

■ Problem 12: The spherical pendulum has Lagrangian, [PU 07]

$$L = \frac{1}{2} m \ell^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) - mg \ell \cos \theta$$

Write down the Euler Lagrange equations of motion and two constants of motion.

Spherical Pendulum

Solution:

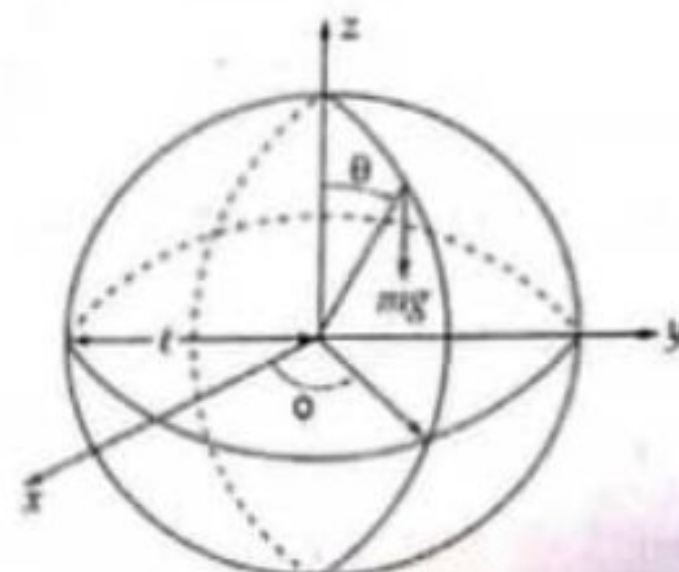


Figure: Spherical pendulum

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CHAPTER - 3: LAGRANGIAN MECHANICS

| 107

Given that

$$L = \frac{1}{2} m \ell^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) - mg \ell \cos \theta$$

Differentiating w.r.t. $\theta, \dot{\theta}$ and $\phi, \dot{\phi}$

$$\frac{\partial L}{\partial \theta} = m \ell^2 \sin \theta \cos \theta \dot{\phi}^2 + mg \ell \sin \theta,$$

$$\frac{\partial L}{\partial \phi} = 0$$

$$\frac{\partial L}{\partial \dot{\theta}} = m \ell^2 \dot{\theta},$$

$$\frac{\partial L}{\partial \dot{\phi}} = m \ell^2 \sin^2 \theta \dot{\phi}$$

Putting in Lagrange's equations

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = 0$$

$$\frac{\partial L}{\partial \phi} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = 0$$

$$m \ell^2 \sin \theta \cos \theta \dot{\phi}^2 + mg \ell \sin \theta - \frac{d}{dt} (m \ell^2 \dot{\theta}) = 0,$$

$$0 - \frac{d}{dt} (m \ell^2 \sin^2 \theta \dot{\phi}) = 0$$

$$m \ell^2 \sin \theta \cos \theta \dot{\phi}^2 + mg \ell \sin \theta - m \ell^2 \ddot{\theta} = 0, \quad -2m \ell^2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} - m \ell^2 \sin^2 \theta \ddot{\phi} = 0$$

$$m \ell^2 \left(\ddot{\theta} - \sin \theta \cos \theta \dot{\phi}^2 + \frac{g}{\ell} \sin \theta \right) = 0, \quad -m \ell^2 \sin^2 \theta \left(2 \frac{\cos \theta}{\sin \theta} \dot{\theta} \dot{\phi} + \ddot{\phi} \right) = 0$$

$$\ddot{\theta} - \sin \theta \cos \theta \dot{\phi}^2 + \frac{g}{\ell} \sin \theta = 0$$

$$\ddot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi} = 0$$

These are equations of motion for a spherical pendulum.



That is, when the integral of motion F does not contain t explicitly, its Poisson bracket with the Hamiltonian of the system vanishes.

Another important property of Poisson brackets is that, if F and G are two integrals of motion, then the Poisson bracket of F and G , $[F, G]$, is also an integral of motion; that is,

$$[F, G] = \text{constant} \quad (5.48)$$

during the motion. This is known as Poisson's theorem.

The proof of the Poisson theorem starts by noting that, because F and G are integrals of motion, we have

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + [F, H] = 0,$$

and

$$\frac{dG}{dt} = \frac{\partial G}{\partial t} + [G, H] = 0$$

or

$$\frac{\partial F}{\partial t} = -[F, H]$$

and

$$\frac{\partial G}{\partial t} = -[G, H]. \quad (5.49)$$

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Hamiltonian Formulation of Mechanics

143

Now, from Equation 5.45, we have

$$\begin{aligned} \frac{d[F, G]}{dt} &= [F, G], H + \frac{\partial[F, G]}{\partial t} \\ &= [F, G], H + \left[\frac{\partial F}{\partial t}, G \right] + \left[F, \frac{\partial G}{\partial t} \right] \end{aligned}$$

which, with the aid of Equation 5.49, becomes

$$\frac{d[F, G]}{dt} = [F, G], H - [F, H], G - [F, [G, H]]$$



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Hamiltonian Formulation of Mechanics

143

Now, from Equation 5.45, we have

$$\begin{aligned}\frac{d[F, G]}{dt} &= [F, G], H + \frac{\partial[F, G]}{\partial t} \\ &= [F, G], H + \left[\frac{\partial F}{\partial t}, G \right] + \left[F, \frac{\partial G}{\partial t} \right]\end{aligned}$$

which, with the aid of Equation 5.49, becomes

$$\begin{aligned}\frac{d[F, G]}{dt} &= [F, G], H - [F, H], G - [F, [G, H]] \\ &= [F, G], H + [H, F], G + [G, H], F = 0.\end{aligned}$$

In the last step, Jacobi's identity was used. Hence, we have

$$[F, G] = \text{constant}.$$

Example 5.5

A particle of mass m is moving in a central potential V that does not depend on velocity. Find the integrals of motion.

Solution:

The kinetic energy of the particle, in spherical coordinates, is

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2)$$

and its Lagrangian function is $L = T - V$. Because V does not depend on V , hence,

$$p_i = \partial L / \partial \dot{q}_i = \partial T / \partial \dot{q}_i$$

from which we obtain

$$p_r = m\dot{r}, \quad p_\theta = mr^2\dot{\theta}, \quad p_\phi = mr^2\sin^2\theta\dot{\phi}.$$

$$f(\theta, \varphi) = r - \rho \sin \theta = 0$$

This constraint equation ensures that the sphere remains in contact with the cylinder.

Lagrange Equations of Motion:

Using the Lagrangian function and the constraint equation, we can write the Lagrange equations of motion as:

$$(d/dt)(\partial L / \partial \dot{\theta}) - (\partial L / \partial \theta) = \lambda (\partial f / \partial \theta)$$

$$(d/dt)(\partial L / \partial \dot{\varphi}) - (\partial L / \partial \varphi) = 0$$

where λ is the Lagrange multiplier.

Small Frequency of Oscillation:

To find the small frequency of oscillation, we can linearize the equations of motion around the equilibrium point $\theta = 0$ (the bottom of the cylinder). After some calculations, we get:

$$\omega^2 = (g/r)$$

This is the small frequency of oscillation of the sphere.

The Lagrangian function for the sphere is given by:

$$L = T - U$$

where T is the kinetic energy and U is the potential energy.

$$T = \frac{1}{2}m(\dot{\theta})^2 + \frac{1}{2}m(\dot{\varphi})^2 \sin^2 \theta$$

$$U = mgh(1 - \cos \theta)$$

where θ is the polar angle, φ is the azimuthal angle, m is the mass, g is the acceleration due to gravity, and h is the radius of the cylinder.

Equation of Constraints:

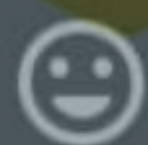
The sphere is constrained to move on the inner surface of the cylinder, so we have:

$$f(\theta, \varphi) = r - \rho \sin \theta = 0$$

This constraint equation ensures that the sphere remains in contact with the cylinder.

Lagrange Equations of Motion:

Using the Lagrangian function and the



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Problem 02

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If L is a Lagrangian for a system of n degrees of freedom satisfying Lagrange's equation of motion, show by direct substitution that

$$L' = L + \frac{d}{dt}F(q_1, q_2, \dots, q_n, t) = L + \dot{F}$$

Also satisfies the Lagrange's equations of motion where F is an arbitrary differentiable function of its argument.

Sol:

Lagrange's Equation for n Degrees of Freedom

If L' is another Lagrangian for a system then Lagrange's equation of motion becomes

$$\frac{d}{dt} \left(\frac{\partial L'}{\partial \dot{q}_i} \right) - \frac{\partial L'}{\partial q_i} = 0$$

Substituting value of L'

$$\frac{d}{dt} \left(\frac{\partial (L + \dot{F})}{\partial \dot{q}_i} \right) - \frac{\partial (L + \dot{F})}{\partial q_i} = 0$$

$$\left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} \right] + \left[\frac{d}{dt} \left(\frac{\partial \dot{F}}{\partial \dot{q}_i} \right) - \frac{\partial \dot{F}}{\partial q_i} \right] = 0$$

As

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

So

$$\frac{d}{dt} \left(\frac{\partial \dot{F}}{\partial \dot{q}_i} \right) - \frac{\partial \dot{F}}{\partial q_i} = 0 \quad \text{----- (1)}$$

Force is given as

$$F = F(q_1, q_2, \dots, q_n, t)$$

$$\frac{dF}{dt} = \frac{\partial F}{\partial q_1} \frac{\partial q_1}{\partial t} + \frac{\partial F}{\partial q_2} \frac{\partial q_2}{\partial t} + \dots + \frac{\partial F}{\partial q_n} \frac{\partial q_n}{\partial t} + \frac{\partial F}{\partial t} \frac{\partial t}{\partial t}$$

$$= \frac{\partial F}{\partial q_1} \dot{q}_1 + \frac{\partial F}{\partial q_2} \dot{q}_2 + \dots + \frac{\partial F}{\partial q_n} \dot{q}_n + \frac{\partial F}{\partial t}$$

$$\dot{F} = \sum_i \frac{\partial F}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial t}$$

$$\frac{\partial \dot{F}}{\partial \dot{q}_i} = \frac{\partial F}{\partial q_i} (1) + 0$$

$$\frac{\partial \dot{F}}{\partial \dot{q}_i} = \frac{\partial F}{\partial q_i}$$

Hence

$$\frac{d}{dt} \left(\frac{\partial \dot{F}}{\partial \dot{q}_i} \right) = \frac{d}{dt} \left(\frac{\partial F}{\partial q_i} \right) = \frac{\partial \dot{F}}{\partial q_i} \quad \text{----- (2)}$$

Which is the same as eq(1). Hence the result is proved.

Problem 02

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6- Jacobi Identity :

$$\{A, \{B, C\}\} + \{B, \{C, A\}\} + \{C, \{A, B\}\} = 0$$

Taking $\{A, \{B, C\}\}$

$$= \frac{\partial A}{\partial x_i} \cdot \frac{\partial \{B, C\}}{\partial p_i} - \frac{\partial A}{\partial p_i} \cdot \frac{\partial \{B, C\}}{\partial x_i}$$

$$= \frac{\partial A}{\partial x_i} \left[\frac{\partial B}{\partial p_i} \left\{ \frac{\partial B}{\partial p_i} \frac{\partial C}{\partial p_i} - \frac{\partial B}{\partial p_i} \frac{\partial C}{\partial x_i} \right\} \right]$$

$$- \frac{\partial A}{\partial p_i} \left[\frac{\partial}{\partial x_i} \left\{ \frac{\partial B}{\partial x_i} \frac{\partial C}{\partial p_i} - \frac{\partial B}{\partial p_i} \frac{\partial C}{\partial x_i} \right\} \right]$$

$$= \frac{\partial A}{\partial x_i} \left[\frac{\partial^2 B}{\partial p_i \partial x_i} \frac{\partial C}{\partial p_i} + \frac{\partial B}{\partial x_i} \frac{\partial^2 C}{\partial p_i^2} - \frac{\partial^2 B}{\partial p_i^2} \frac{\partial C}{\partial x_i} - \frac{\partial B}{\partial p_i} \frac{\partial^2 C}{\partial p_i \partial x_i} \right]$$

$$- \frac{\partial A}{\partial p_i} \left[\frac{\partial^2 B}{\partial x_i^2} \frac{\partial C}{\partial p_i} + \frac{\partial B}{\partial x_i} \frac{\partial^2 C}{\partial x_i \partial p_i} - \frac{\partial^2 B}{\partial p_i \partial x_i} \frac{\partial C}{\partial x_i} - \frac{\partial B}{\partial p_i} \frac{\partial^2 C}{\partial x_i^2} \right]$$

$$= \frac{\partial A}{\partial x_i} \frac{\partial^2 B}{\partial p_i \partial x_i} \frac{\partial C}{\partial p_i} + \frac{\partial A}{\partial x_i} \frac{\partial B}{\partial x_i} \frac{\partial^2 C}{\partial p_i^2} - \frac{\partial A}{\partial x_i} \frac{\partial^2 B}{\partial p_i^2} \frac{\partial C}{\partial x_i}$$

$$- \frac{\partial A}{\partial x_i} \frac{\partial B}{\partial p_i} \frac{\partial^2 C}{\partial p_i \partial x_i} - \frac{\partial A}{\partial p_i} \frac{\partial^2 B}{\partial x_i^2} \frac{\partial C}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial x_i} \frac{\partial^2 C}{\partial x_i \partial p_i}$$

$$+ \frac{\partial A}{\partial p_i} \frac{\partial^2 B}{\partial p_i \partial x_i} \frac{\partial C}{\partial x_i} + \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial p_i} \frac{\partial^2 C}{\partial x_i^2} \quad (23)$$

Now taking $\{B, \{C, A\}\}$

Replace (A, B, C) by (B, C, A) in (23)

$$\{B, \{C, A\}\} = \frac{\partial B}{\partial x_i} \frac{\partial^2 C}{\partial p_i \partial x_i} \frac{\partial A}{\partial p_i} + \frac{\partial B}{\partial x_i} \frac{\partial C}{\partial x_i} \frac{\partial^2 A}{\partial p_i^2}$$

$$- \frac{\partial B}{\partial x_i} \frac{\partial^2 C}{\partial p_i^2} \frac{\partial A}{\partial x_i} - \frac{\partial B}{\partial x_i} \frac{\partial C}{\partial p_i} \frac{\partial^2 A}{\partial p_i \partial x_i} - \frac{\partial B}{\partial p_i} \frac{\partial^2 C}{\partial x_i^2} \frac{\partial A}{\partial p_i}$$

(17)

$$\begin{aligned}
 &= \frac{\partial B}{\partial x_i} \cdot \frac{\partial C}{\partial x_j} \cdot \frac{\partial^2 A}{\partial x_i \partial x_j} + \frac{\partial B}{\partial x_i} \cdot \frac{\partial^2 C}{\partial x_i \partial x_j} \cdot \frac{\partial A}{\partial x_j} \\
 &+ \frac{\partial B}{\partial x_i} \cdot \frac{\partial C}{\partial x_j} \cdot \frac{\partial^2 A}{\partial x_i^2} \quad \dots \quad (24)
 \end{aligned}$$

Now taking $\{C, \{A, B\}\}$

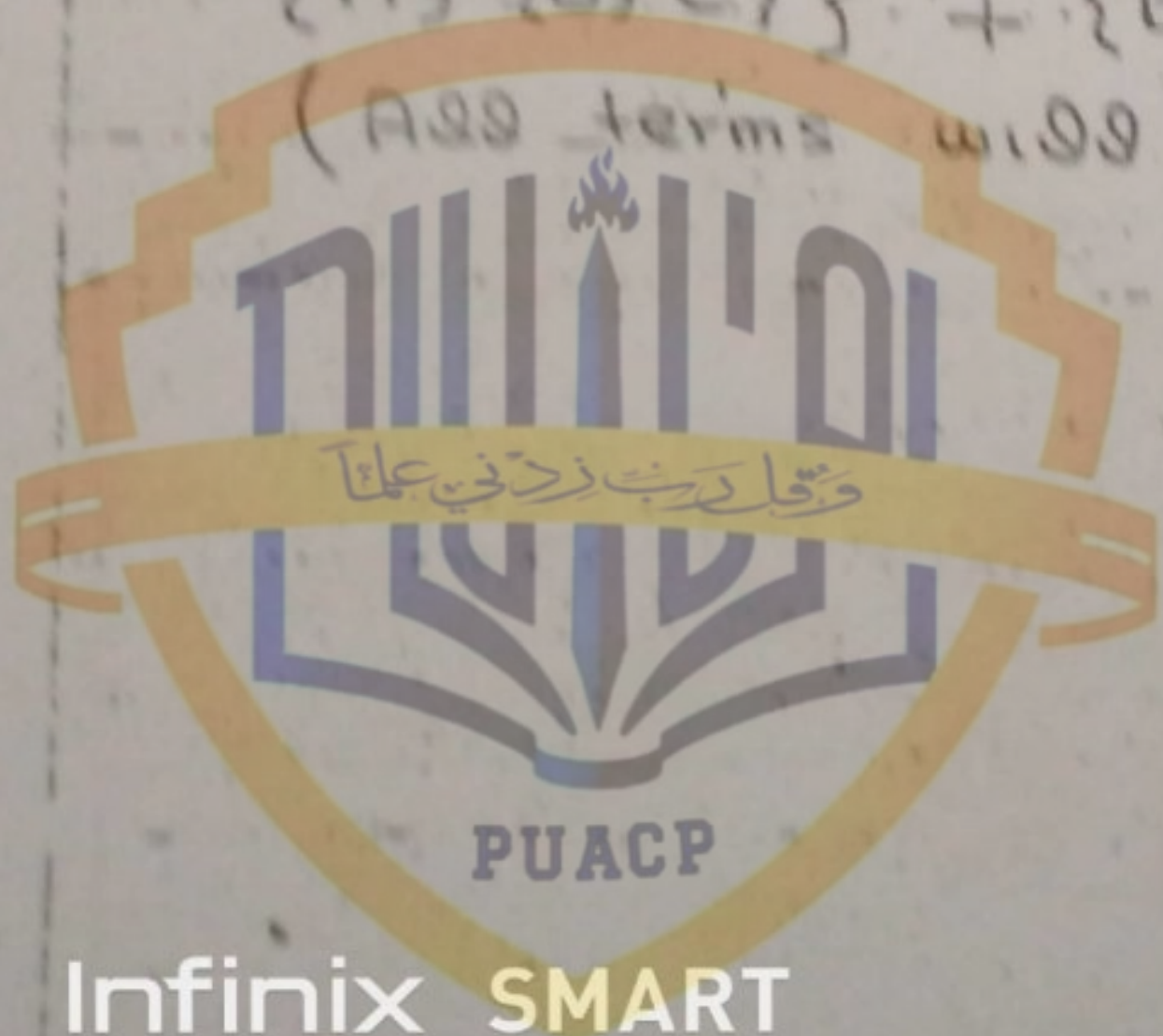
Replace $(A, B, C) = (B, C, A)$ in (24)

$$\begin{aligned}
 &= \frac{\partial C}{\partial x_i} \cdot \frac{\partial^2 A}{\partial x_i \partial x_j} \cdot \frac{\partial B}{\partial x_j} - \frac{\partial A}{\partial x_i} \cdot \frac{\partial^2 B}{\partial x_i^2} \cdot \frac{\partial C}{\partial x_j} - \frac{\partial C}{\partial x_i} \cdot \frac{\partial^2 A}{\partial x_i^2} \cdot \frac{\partial B}{\partial x_j} \\
 &- \frac{\partial C}{\partial x_i} \cdot \frac{\partial A}{\partial x_j} \cdot \frac{\partial^2 B}{\partial x_i \partial x_j} - \frac{\partial C}{\partial x_i} \cdot \frac{\partial^2 A}{\partial x_i^2} \cdot \frac{\partial B}{\partial x_j} - \frac{\partial C}{\partial x_i} \cdot \frac{\partial A}{\partial x_j} \cdot \frac{\partial^2 B}{\partial x_i \partial x_j} \\
 &+ \frac{\partial C}{\partial x_i} \cdot \frac{\partial^2 A}{\partial x_i \partial x_j} \cdot \frac{\partial B}{\partial x_j} + \frac{\partial C}{\partial x_i} \cdot \frac{\partial A}{\partial x_j} \cdot \frac{\partial^2 B}{\partial x_i^2} \quad \dots \quad (25)
 \end{aligned}$$

Now adding (23), (24) and (25) we have

$$\{A, \{B, C\}\} + \{B, \{C, A\}\} + \{C, \{A, B\}\} = 0$$

(All terms will cancel with each other)



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The most important type of central force is the one in which the force varies inversely as the square of the radial distance:

$$f(r) = -\frac{k}{r^2} \quad \text{or} \quad V(r) = -\frac{k}{r}$$

where the constant k is positive for an attractive force (e.g., $k = Gm_1m_2$ for attractive gravitational force), and k is negative for a repulsive force.

The equation of the orbit can be obtained from Equation 6.31. The advantage of doing this is to avoid the complicated integration. For inverse square force $f(1/u) = -ku^2$, Equation 6.31 becomes

$$\frac{d^2u}{d\theta^2} + u = \frac{\mu k}{l^2}$$

which can be rewritten as

$$\frac{d^2(u - \mu k/l^2)}{d\theta^2} + (u - \mu k/l^2) = 0.$$

Its solution is

$$u - \mu k/l^2 = b \cos(\theta - \theta_0)$$

or

$$r = \frac{\alpha}{1 + p \cos(\theta - \theta_0)} \quad (6.34)$$

where

$$\alpha = \frac{l^2}{\mu k}, \quad p = \frac{bl^2}{\mu k}$$

and b and θ_0 are integration constants.

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Equation 6.34 is the equation of a conic section in polar coordinates. By definition, a conic section in polar coordinate representation is the curve traced by a point P (Figure 6.8) that moves so that the ratio of its distance from a fixed point F (the focus) to its distance from a straight line D (the directrix) is constant, denoted by ϵ , called the eccentricity of the cone. A conic section is an ellipse, parabola, or hyperbola according to whether $\epsilon < 1$, $\epsilon = 1$, or $\epsilon > 1$. Appendix 4 is prepared for readers who need to review conic sections.

If r is the distance from the focus to P , and d is the distance of the focus from the directrix, then

$$\epsilon = \frac{r}{d - r \cos \theta} \quad \text{or} \quad r = \frac{d\epsilon}{1 + \epsilon \cos \theta}. \quad (6.35)$$

Comparing this with Equation 6.34, we see that the path of the particle is a conic section of



Comparing this with Equation 6.34, we see that the path of the particle is a conic section of eccentricity

$$\epsilon = p = \frac{bl^2}{\mu k}.$$

Equation 6.34 can also be obtained from Equation 7.25, which now takes the form

$$\theta = \theta_0 - \int \frac{du}{[(2\mu E/l^2 + (2\mu k u/l^2) - u^2)]^{1/2}} \quad (6.36)$$

where θ_0 is a constant of integration determined by the initial condition. The integration on the right-hand side is of the standard form:

$$\int \frac{dx}{(a + bx + cx^2)^{1/2}} = \frac{1}{\sqrt{-c}} \cos^{-1} \left(-\frac{b + 2cx}{\sqrt{q}} \right)$$

where $q = b^2 - 4ac$. To apply this to Equation 6.35, we set

$$a = 2\mu E/l^2, \quad b = 2\mu k/l^2, \quad c = -1, \quad q = \left(\frac{2\mu k}{l^2} \right)^2 \left(1 + \frac{2El^2}{\mu k^2} \right)$$

and obtain

$$\frac{1}{r} = \frac{\mu k}{l^2} \left[1 + \sqrt{1 + \frac{2El^2}{\mu k}} \cos(\theta - \theta_0) \right]$$

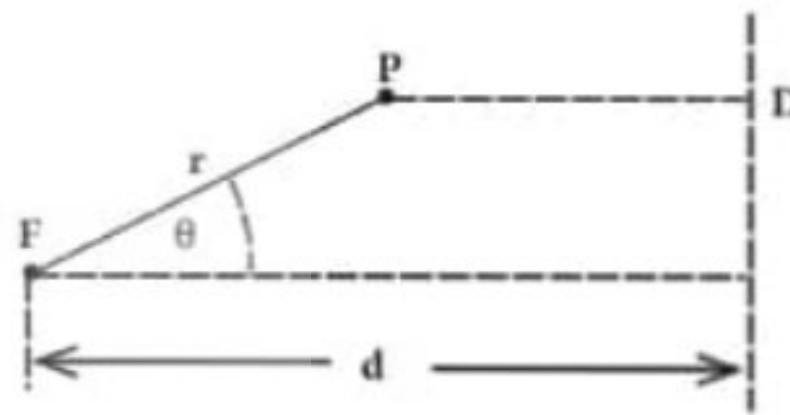


FIGURE 6.8 Conic sections.

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Motion Under a Central Force

or
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169

$$r = \frac{\alpha}{1 + \beta \cos(\theta - \theta_0)} \quad (6.37)$$

where



or

$$r = \frac{\alpha}{1 + \beta \cos(\theta - \theta_0)} \quad (6.37)$$

where

$$\alpha = \frac{l^2}{\mu k} \quad \text{and} \quad \beta = \sqrt{1 + \frac{2El^2}{\mu k^2}}. \quad (6.38)$$

Comparing this with Equation 6.35, we see that Equation 6.37 is a conic section of eccentricity

$$\varepsilon = \beta = \sqrt{1 + \frac{2El^2}{\mu k^2}}. \quad (6.38a)$$

It remains to be shown that Equation 6.37 is identical to Equation 6.34. If we can show that p and β are identical in these two equations, then the two equations are also identical. To this end, we start with Equation 6.17, which now takes the form

$$\frac{1}{2} \mu \dot{r}^2 + \frac{l^2}{2\mu r^2} - \frac{k}{r} = E.$$

When r has its minimum value $r_{\min}$, the particle is turning so that $\dot{r} = 0$, and the energy equation reduces to

$$E = \frac{l^2}{2\mu r_{\min}^2} - \frac{k}{r_{\min}}.$$

Now, from Equation 6.34, we have, by setting $\cos(\theta - \theta_0) = 1$,

$$r_{\min} = \alpha/(1 + p).$$

Substituting this into the preceding equation for E and solving for p , we obtain

$$p = \sqrt{1 + \frac{2El^2}{\mu k^2}} = \beta,$$

and Equation 6.34 is identical to Equation 6.37.

The value of θ_0 merely determines the orientation of the orbit, so without loss of generality in discussing the form of the orbit, we can choose $\theta_0 = 0$. Then, the equation of the orbit (Equation 6.37) becomes

$$r = \frac{\alpha}{1 + \varepsilon \cos \theta} \quad (6.39)$$

where α and ε are given by Equation 6.38.

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so given transformation is canonical differential

PROVE that the transformation is canonical.

and $P = q \cot p$ is canonical. $Q = \log\left(\frac{1}{q} \sin p\right)$

$$\text{Taking } pdq - P dQ = pdq - (q \cot p) \cdot \frac{1}{\frac{\sin p}{q}} \cdot d\left(\frac{\sin p}{q}\right)$$

$$= pdq - \frac{q^2 \cot p}{\sin p} \cdot d\left(\frac{\sin p}{q}\right)$$

$$= pdq - \frac{q^2 \cos p}{\sin^2 p} \left(\frac{q \cdot \cos p dp - \sin p \cdot dq}{q^2} \right)$$

$$= pdq - \frac{\cos p}{\sin^2 p} (q \cos p dp - \sin p dq)$$

$$= pdq - \frac{q \cos^2 p}{\sin^2 p} dp + \frac{\cos p}{\sin p} dq$$

$$= \left(p + \frac{\cos p}{\sin p} \right) dq - q \cot^2 p dp$$

$$= \left(p + \frac{\cos p}{\sin p} \right) dq - q (\operatorname{cosec}^2 p - 1) dp$$

$$= (p + \cot p) dq + q(1 - \operatorname{cosec}^2 p) dp$$

$$= d \left\{ q(p + \cot p) \right\}$$

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